

B2: Symmetry and Relativity

Model Solutions — Trinity Term 2024

Conventions. Metric $\eta_{\mu\nu} = \text{diag}(+, -, -, -)$. $U^\mu = (\gamma c, \gamma \mathbf{v})$, $U_\mu U^\mu = c^2$. $p^\mu = (E/c, \mathbf{p})$, $p_\mu p^\mu = m^2 c^2$. SI units; $c = 2.998 \times 10^8 \text{ m s}^{-1}$, $M_e c^2 = 0.511 \text{ MeV}$.

Question 1 — Photon–photon pair production, helicity, charge conjugation, and Compton/annihilation kinematics

Problem. (a) Threshold E_γ for $\gamma + \gamma_{\text{CMB}} \rightarrow e^+ e^-$ with $\varepsilon_{\text{CMB}} = 6.626 \times 10^{-4} \text{ eV}$. (b) Boost flipping helicity of an electron (M_e, p); massless case. (c) Show $\tau \rightarrow -\tau$ in the Lorentz force law gives $q \rightarrow -q$; rewrite the scalar $\mathcal{S}$ for a positron. (d) Compton: $\lambda_2 - \lambda_1 = (h/M_e c)(1 - \cos \theta)$. (e) Two-photon annihilation: $\lambda_1 + \lambda_2 = (h/M_e c)(1 - \cos \theta)$.

(a) Threshold for $\gamma + \gamma_{\text{CMB}} \rightarrow e^+ e^-$

At threshold the pair is at rest in the CoM:

$$s_{\text{min}} = (2M_e c)^2 = 4M_e^2 c^2.$$

Head-on geometry minimises E_γ :

$$p_1^\mu = (E_\gamma/c)(1, 1, 0, 0), \quad p_2^\mu = (\varepsilon/c)(1, -1, 0, 0).$$

Compute the dot product with metric $(+, -, -, -)$:

$$p_1 \cdot p_2 = \frac{E_\gamma}{c} \cdot \frac{\varepsilon}{c} - \frac{E_\gamma}{c} \cdot \left(-\frac{\varepsilon}{c}\right).$$

Combine the two terms:

$$p_1 \cdot p_2 = \frac{2E_\gamma \varepsilon}{c^2}.$$

Form s using $p_1^2 = p_2^2 = 0$:

$$s = (p_1 + p_2)^2 = 2p_1 \cdot p_2 = \frac{4E_\gamma \varepsilon}{c^2}.$$

Equate to threshold $4M_e^2 c^2$:

$$\frac{4E_\gamma \varepsilon}{c^2} = 4M_e^2 c^2.$$

Solve for E_γ :

$$E_\gamma^{\text{thr}} = \frac{(M_e c^2)^2}{\varepsilon_{\text{CMB}}}.$$

Insert numbers:

$$E_\gamma^{\text{thr}} = \frac{(5.11 \times 10^5 \text{ eV})^2}{6.626 \times 10^{-4} \text{ eV}}.$$

Reduce:

$$E_{\gamma}^{\text{thr}} = \frac{2.611 \times 10^{11} \text{ eV}^2}{6.626 \times 10^{-4} \text{ eV}} = 3.94 \times 10^{14} \text{ eV}.$$

$$E_{\gamma}^{\text{thr}} = \frac{(M_e c^2)^2}{\varepsilon_{\text{CMB}}} \approx 3.9 \times 10^{14} \text{ eV} \approx 400 \text{ TeV}. \quad (1)$$

(b) Helicity flip by boost

Helicity $h = \hat{\mathbf{s}} \cdot \hat{\mathbf{p}}$. A collinear boost cannot rotate $\hat{\mathbf{s}}$, so flipping h requires reversing $\mathbf{p}$. Boost along $+\hat{\mathbf{p}}$ at β :

$$p'_{\parallel} = \gamma(p - \beta E/c).$$

Require $p'_{\parallel} < 0$:

$$p - \beta E/c < 0.$$

Rearrange:

$$\beta E/c > p.$$

Solve for β :

$$\beta > \frac{pc}{E}.$$

Use the mass-shell $E = \sqrt{p^2 c^2 + M_e^2 c^4}$ in the threshold value:

$$\vec{\beta} = \frac{pc}{E} \hat{\mathbf{p}}, \quad E = \sqrt{p^2 c^2 + M_e^2 c^4}; \text{ any larger } \beta \text{ along } +\hat{\mathbf{p}} \text{ flips helicity.} \quad (2)$$

Massless case: $E = pc$ requires $\beta = 1$, not a Lorentz transformation. Hence helicity is Lorentz invariant for a photon and *no frame* flips it.

(c) Charge conjugation via $\tau \rightarrow -\tau$ and the positron scalar

Lorentz force law:

$$m \frac{d^2 x^{\alpha}}{d\tau^2} = q F^{\alpha\beta} U_{\beta}, \quad U_{\beta} = \frac{dx_{\beta}}{d\tau}. \quad (3)$$

Under $\tau \rightarrow -\tau$ the second derivative is unchanged:

$$\frac{d^2 x^{\alpha}}{d(-\tau)^2} = \frac{d^2 x^{\alpha}}{d\tau^2}.$$

The 4-velocity picks up a sign:

$$U_{\beta} \rightarrow \frac{dx_{\beta}}{d(-\tau)} = -U_{\beta}.$$

Substitute into the force law:

$$m \frac{d^2 x^{\alpha}}{d\tau^2} = q F^{\alpha\beta} (-U_{\beta}) = (-q) F^{\alpha\beta} U_{\beta}.$$

This is the equation for charge $-q$. τ is a Lorentz scalar, so the statement is frame-independent.

Stueckelberg–Feynman dictionary (positron = time-reversed electron):

$$p_2^i \rightarrow -p_{e+}^f, \quad p_2^f \rightarrow -p_{e+}^i.$$

Substitute into each numerator bilinear:

$$(p_2^f \cdot p_1^f)(p_1^i \cdot p_2^i) \rightarrow (-p_{e+}^i \cdot p_1^f)(-p_1^i \cdot p_{e+}^f) = (p_{e+}^i \cdot p_1^f)(p_1^i \cdot p_{e+}^f).$$

The second numerator term:

$$(p_2^f \cdot p_2^i)(p_1^i \cdot p_1^f) \rightarrow ((-p_{e+}^i) \cdot (-p_{e+}^f))(p_1^i \cdot p_1^f).$$

Cancel the two minus signs:

$$(p_2^f \cdot p_2^i)(p_1^i \cdot p_1^f) \rightarrow (p_{e+}^i \cdot p_{e+}^f)(p_1^i \cdot p_1^f).$$

The denominator:

$$(p_2^f - p_1^i)^4 \rightarrow (-p_{e+}^i - p_1^i)^4 = (p_{e+}^i + p_1^i)^4.$$

Collect:

$$\boxed{\mathcal{S}_{ee+} = \frac{(p_{e+}^i \cdot p_1^f)(p_1^i \cdot p_{e+}^f) + (p_{e+}^i \cdot p_{e+}^f)(p_1^i \cdot p_1^f)}{(p_{e+}^i + p_1^i)^4}}. \quad (4)$$

The denominator becomes the squared CoM energy $s = (p_{e+}^i + p_1^i)^2$: t -channel Møller becomes s -channel Bhabha by crossing.

(d) Compton scattering

4-momentum conservation:

$$p_\gamma^i + p_e^i = p_\gamma^f + p_e^f.$$

Isolate the final electron:

$$p_e^f = p_\gamma^i + p_e^i - p_\gamma^f.$$

Square both sides:

$$(p_e^f)^2 = (p_\gamma^i + p_e^i - p_\gamma^f)^2.$$

Expand the right-hand side:

$$(p_e^f)^2 = (p_\gamma^i)^2 + (p_e^i)^2 + (p_\gamma^f)^2 + 2p_\gamma^i \cdot p_e^i - 2p_\gamma^f \cdot p_e^i - 2p_\gamma^i \cdot p_\gamma^f.$$

Apply mass shells $(p_e^{if})^2 = M_e^2 c^2$, $(p_\gamma^{if})^2 = 0$:

$$0 = 2p_\gamma^i \cdot p_e^i - 2p_\gamma^f \cdot p_e^i - 2p_\gamma^i \cdot p_\gamma^f.$$

Divide by 2:

$$p_\gamma^i \cdot p_e^i - p_\gamma^f \cdot p_e^i = p_\gamma^i \cdot p_\gamma^f.$$

Evaluate the first dot product in lab frame ($p_e^i = (M_e c, \vec{0})$):

$$p_\gamma^i \cdot p_e^i = (E_1/c)(M_e c) = M_e E_1.$$

Similarly for the second:

$$p_\gamma^f \cdot p_e^i = (E_2/c)(M_e c) = M_e E_2.$$

Photon-photon dot product with $|\vec{p}_{1,2}| = E_{1,2}/c$ and opening angle θ :

$$p_\gamma^i \cdot p_\gamma^f = \frac{E_1 E_2}{c^2} - \frac{E_1 E_2}{c^2} \cos \theta.$$

Factor:

$$p_\gamma^i \cdot p_\gamma^f = \frac{E_1 E_2}{c^2} (1 - \cos \theta).$$

Substitute all three into the divided equation:

$$M_e E_1 - M_e E_2 = \frac{E_1 E_2}{c^2} (1 - \cos \theta).$$

Factor M_e :

$$M_e (E_1 - E_2) = \frac{E_1 E_2}{c^2} (1 - \cos \theta).$$

Divide by $M_e E_1 E_2$:

$$\frac{1}{E_2} - \frac{1}{E_1} = \frac{1 - \cos \theta}{M_e c^2}.$$

Use $E = hc/\lambda$:

$$\frac{\lambda_2}{hc} - \frac{\lambda_1}{hc} = \frac{1 - \cos \theta}{M_e c^2}.$$

Multiply by hc :

$$\boxed{\lambda_2 - \lambda_1 = \frac{h}{M_e c} (1 - \cos \theta).} \quad (5)$$

Since $1 - \cos \theta \geq 0$, $\lambda_2 \geq \lambda_1$: the *incident* photon (1) is shorter.

(e) Two-photon annihilation

Process $e^-(p_e) + e^+(p_{e+}) \rightarrow \gamma_1(k_1) + \gamma_2(k_2)$, positron at rest. Conservation:

$$p_e + p_{e+} = k_1 + k_2.$$

Isolate the electron:

$$p_e = k_1 + k_2 - p_{e+}.$$

Square both sides:

$$p_e^2 = (k_1 + k_2 - p_{e+})^2.$$

Expand the right-hand side:

$$p_e^2 = k_1^2 + k_2^2 + p_{e+}^2 + 2k_1 \cdot k_2 - 2k_1 \cdot p_{e+} - 2k_2 \cdot p_{e+}.$$

Apply mass shells $p_e^2 = p_{e+}^2 = M_e^2 c^2$, $k_1^2 = k_2^2 = 0$:

$$0 = 2k_1 \cdot k_2 - 2k_1 \cdot p_{e+} - 2k_2 \cdot p_{e+}.$$

Divide by 2 and rearrange:

$$k_1 \cdot k_2 = k_1 \cdot p_{e+} + k_2 \cdot p_{e+}.$$

Evaluate dot products in positron rest frame ($p_{e+} = (M_e c, \vec{0})$):

$$k_i \cdot p_{e+} = (E_i/c)(M_e c) = M_e E_i.$$

Photon–photon dot product with opening angle θ :

$$k_1 \cdot k_2 = \frac{E_1 E_2}{c^2} - \frac{E_1 E_2}{c^2} \cos \theta = \frac{E_1 E_2}{c^2} (1 - \cos \theta).$$

Substitute into the rearranged conservation equation:

$$\frac{E_1 E_2}{c^2} (1 - \cos \theta) = M_e E_1 + M_e E_2.$$

Factor M_e on the right:

$$\frac{E_1 E_2}{c^2} (1 - \cos \theta) = M_e (E_1 + E_2).$$

Divide by $M_e E_1 E_2$:

$$\frac{1 - \cos \theta}{M_e c^2} = \frac{1}{E_1} + \frac{1}{E_2}.$$

Substitute $E_i = hc/\lambda_i$:

$$\frac{1 - \cos \theta}{M_e c^2} = \frac{\lambda_1}{hc} + \frac{\lambda_2}{hc}.$$

Multiply by hc :

$$\lambda_1 + \lambda_2 = \frac{h}{M_e c} (1 - \cos \theta). \quad (6)$$

Sign change vs. (5) reflects the crossing $k \rightarrow -k$. At $\theta = \pi$ with $\vec{p}_e = \vec{0}$: $\lambda_1 = \lambda_2 = h/(M_e c)$, two back-to-back 511 keV gammas.

Question 2 — Charged-particle motion in $F^{\mu\nu}$, gauge invariance, plane waves and field invariants

Problem. (a) Pure $\vec{B}$: show $\vec{F} \perp \vec{v}$ and find Larmor radius. (b) From $\partial_\nu \partial^\nu A^\mu - \partial^\mu \partial_\nu A^\nu = -\mu_0 J^\mu$ with static A^0 and $\vec{A} = 0$, recover Poisson and identify J^μ . (c) Plane wave $A^\mu = \epsilon^\mu \sin(k \cdot x)$: derive F^{0i}, F^{ij} and source-free condition. (d) Dual tensor and invariants; show $\vec{E} \perp \vec{B}$, $|\vec{B}| = |\vec{E}|/c$ gauge-free. (e) Find B_1, B_2 from invariants.

(a) Pure magnetic field

Covariant equation:

$$m \frac{dU^\alpha}{d\tau} = q F^{\alpha\beta} U_\beta. \quad (7)$$

With $\vec{E} = 0$ only the spatial components of $F^{\alpha\beta}$ survive. Take $\alpha = 0$:

$$m \frac{d(\gamma c)}{d\tau} = q F^{0\beta} U_\beta = 0.$$

Hence γ and $|\vec{v}|$ are constant. Spatial component, using $d/d\tau = \gamma d/dt$:

$$\frac{d(\gamma m \vec{v})}{dt} = q \vec{v} \times \vec{B}.$$

Since γ is constant the 3-force is:

$$\vec{F} = \gamma m \frac{d\vec{v}}{dt} = q \vec{v} \times \vec{B}.$$

Dot with $\vec{v}$:

$$\vec{v} \cdot \vec{F} = q \vec{v} \cdot (\vec{v} \times \vec{B}) = 0.$$

So $\vec{F} \perp \vec{v}$. Take $\vec{v} \perp \vec{B}$ so the trajectory lies in a plane; force magnitude:

$$|\vec{F}| = |q| v B.$$

Equate to centripetal $|\vec{F}| = \gamma m v^2 / r$ (valid because γ is constant):

$$\frac{\gamma m v^2}{r} = q v B.$$

Cancel one factor of v :

$$\frac{\gamma m v}{r} = q B.$$

Solve for r :

$$\boxed{r = \frac{\gamma m v}{q B} = \frac{p}{q B}.} \quad (8)$$

Assumptions: (i) $\vec{E} = 0$; (ii) $\vec{v} \perp \vec{B}$ (otherwise a helix of the same radius); (iii) γ constant.

(b) Recovering Poisson

$$\partial_\nu \partial^\nu A^\mu - \partial^\mu (\partial_\nu A^\nu) = -\mu_0 J^\mu. \quad (9)$$

Set $\vec{A} = 0$ and $A^0 = A^0(\vec{x})$ (time-independent). The gauge term:

$$\partial_\nu A^\nu = \partial_0 A^0 + \partial_i A^i = 0 + 0 = 0.$$

Take $\mu = 0$. Static A^0 removes the time derivative from $\partial_\nu \partial^\nu$:

$$\partial_\nu \partial^\nu A^0 = \nabla^2 A^0.$$

Substitute into (9) with $J^0 = c\rho$ and the gauge term vanishing:

$$\nabla^2 A^0 = -\mu_0 c\rho.$$

Use $A^0 = \phi/c$:

$$\frac{1}{c} \nabla^2 \phi = -\mu_0 c\rho.$$

Multiply through by c :

$$\nabla^2 \phi = -\mu_0 c^2 \rho.$$

Use $\mu_0 c^2 = 1/\epsilon_0$:

$$\boxed{\nabla^2 \phi = -\frac{\rho}{\epsilon_0}.} \quad (10)$$

Take $\mu = i$ in (9). With $\vec{A} = 0$ and $\partial_\nu A^\nu = 0$:

$$0 - 0 = -\mu_0 J^i.$$

Solve:

$$J^i = 0.$$

Hence

$$\boxed{J^\mu = (c\rho, \vec{0}).} \quad (11)$$

No gauge was fixed: time-independence of A^0 alone killed the gauge term.

(c) Plane-wave field tensor

Differentiate $A^\mu = \epsilon^\mu \sin(k \cdot x)$:

$$\partial^\mu A^\nu = \epsilon^\nu k^\mu \cos(k \cdot x).$$

Form the field tensor:

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu = (k^\mu \epsilon^\nu - k^\nu \epsilon^\mu) \cos(k \cdot x). \quad (12)$$

Compute F^{0i} with $\partial^0 = \partial_0 = (1/c)\partial_t$ and $\partial^i = -\partial_i$:

$$F^{0i} = \partial^0 A^i - \partial^i A^0 = \frac{1}{c} \partial_t A^i + \partial_i(\phi/c).$$

Factor $1/c$:

$$F^{0i} = \frac{1}{c} (\partial_t A^i + \partial_i \phi).$$

Use $E^i = -\partial_t A^i - \partial_i \phi$:

$$F^{0i} = -\frac{E^i}{c}.$$

For the spatial-spatial block use $\vec{B} = \nabla \times \vec{A}$, so $(\nabla \times \vec{A})^k = \varepsilon^{kij} \partial_i A^j$:

$$\partial_i A^j - \partial_j A^i = \varepsilon^{ijk} B_k.$$

Identify with $F^{ij} = \partial^i A^j - \partial^j A^i$ (sign convention of the problem):

$$\boxed{F^{0i} = -\frac{E^i}{c}, \quad F^{ij} = \varepsilon^{ijk} B_k.} \quad (13)$$

Source-free Maxwell: $\partial_\mu F^{\mu\nu} = 0$. Differentiate (12):

$$\partial_\mu F^{\mu\nu} = -(k_\mu k^\mu \epsilon^\nu - k^\nu k_\mu \epsilon^\mu) \sin(k \cdot x).$$

Collect:

$$\partial_\mu F^{\mu\nu} = -[k^2 \epsilon^\nu - k^\nu (k \cdot \epsilon)] \sin(k \cdot x).$$

Vanishing for all ν requires

$$k^2 = 0 \quad \text{and} \quad k \cdot \epsilon = 0.$$

The first is the lightlike dispersion; the second is forced transversality (not a gauge choice). The trivial $\epsilon^\nu \propto k^\nu$ is pure gauge with $F^{\mu\nu} = 0$.

(d) Dual tensor, invariants, $\vec{E} \perp \vec{B}$

Definition:

$$\tilde{F}^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\alpha\beta} F_{\alpha\beta}.$$

Component computation (with $\varepsilon_{0123} = +1$, so $\varepsilon^{0123} = -1$ in metric $(+, -, -, -)$):

$$\tilde{F}^{\mu\nu} = \begin{pmatrix} 0 & -B_x & -B_y & -B_z \\ B_x & 0 & E_z/c & -E_y/c \\ B_y & -E_z/c & 0 & E_x/c \\ B_z & E_y/c & -E_x/c & 0 \end{pmatrix}. \quad (14)$$

Two field invariants:

$$F_{\mu\nu} F^{\mu\nu} = 2(\vec{B}^2 - \vec{E}^2/c^2), \quad F_{\mu\nu} \tilde{F}^{\mu\nu} \propto \vec{E} \cdot \vec{B}.$$

Apply (12) to compute the first invariant:

$$F^{\mu\nu}F_{\mu\nu} = (k^\mu\epsilon^\nu - k^\nu\epsilon^\mu)(k_\mu\epsilon_\nu - k_\nu\epsilon_\mu)\cos^2(k \cdot x).$$

Expand the product term by term:

$$F^{\mu\nu}F_{\mu\nu} = [(k \cdot k)(\epsilon \cdot \epsilon) - (k \cdot \epsilon)^2 - (k \cdot \epsilon)^2 + (k \cdot k)(\epsilon \cdot \epsilon)]\cos^2(k \cdot x).$$

Combine like terms:

$$F_{\mu\nu}F^{\mu\nu} = 2[k^2\epsilon^2 - (k \cdot \epsilon)^2]\cos^2(k \cdot x).$$

Compute the dual contraction:

$$\varepsilon_{\mu\nu\alpha\beta}F^{\mu\nu}F^{\alpha\beta} \propto \varepsilon_{\mu\nu\alpha\beta}(k^\mu\epsilon^\nu - k^\nu\epsilon^\mu)(k^\alpha\epsilon^\beta - k^\beta\epsilon^\alpha).$$

Each term has two k 's and two ϵ 's contracted with the totally antisymmetric tensor:

$$\varepsilon_{\mu\nu\alpha\beta}k^\mu k^\alpha = 0, \quad \varepsilon_{\mu\nu\alpha\beta}\epsilon^\nu \epsilon^\beta = 0.$$

Hence

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = 0 \Rightarrow \vec{E} \cdot \vec{B} = 0.$$

$$\boxed{\vec{E} \cdot \vec{B} = 0} \quad (\text{gauge-free}). \quad (15)$$

Apply $k^2 = 0$, $k \cdot \epsilon = 0$ from (c) to the first invariant:

$$F_{\mu\nu}F^{\mu\nu} = 2[0 - 0]\cos^2(k \cdot x) = 0.$$

Equate to $2(\vec{B}^2 - \vec{E}^2/c^2)$:

$$\vec{B}^2 - \vec{E}^2/c^2 = 0.$$

Solve:

$$\boxed{|\vec{B}| = \frac{|\vec{E}|}{c}}. \quad (16)$$

Both invariants vanish: a free plane wave is a null EM field.

(e) Finding B_1, B_2

Two invariants (from (d)): $\vec{E} \cdot \vec{B}$ and $|\vec{E}|^2 - c^2|\vec{B}|^2$. Frame S : $\vec{E} = (E, E, 0)$, $\vec{B} = (0, 0, 2E/c)$. Compute the first invariant in S :

$$\vec{E} \cdot \vec{B} = E \cdot 0 + E \cdot 0 + 0 \cdot (2E/c) = 0.$$

Compute the second invariant in S :

$$|\vec{E}|^2 - c^2|\vec{B}|^2 = (E^2 + E^2) - c^2 \cdot (2E/c)^2 = 2E^2 - 4E^2 = -2E^2.$$

Frame S' : $\vec{E}' = (0, 0, E)$, $\vec{B}' = (B_1, E/c, B_2)$. Compute the first invariant in S' :

$$\vec{E}' \cdot \vec{B}' = 0 \cdot B_1 + 0 \cdot (E/c) + E \cdot B_2 = EB_2.$$

Equate to S :

$$EB_2 = 0 \Rightarrow \boxed{B_2 = 0}.$$

Compute the second invariant in S' (with $B_2 = 0$):

$$|\vec{E}'|^2 - c^2|\vec{B}'|^2 = E^2 - c^2(B_1^2 + E^2/c^2).$$

Expand:

$$|\vec{E}'|^2 - c^2|\vec{B}'|^2 = E^2 - c^2B_1^2 - E^2 = -c^2B_1^2.$$

Equate to S :

$$-c^2B_1^2 = -2E^2.$$

Divide by $-c^2$:

$$B_1^2 = \frac{2E^2}{c^2}.$$

Take square root:

$$\boxed{B_1 = \pm \frac{\sqrt{2} E}{c}.} \quad (17)$$

Sign ambiguity: the invariants do not fix the boost orientation.

Question 3 — Length contraction, the wavefront invariant, beaming and aberration

Problem. (a) 3–4–5 triangle in S' with $3L$ along $\hat{\mathbf{x}}'$, S' boosted at v : find sides in S , area in S' , hypotenuse angle in S . (b) Show L^2 between two points on a wavefront is Lorentz invariant. (c) Show $I'/I = (\omega'/\omega)^2$. (d) Aberration formula and $d\Omega/d\Omega' = (\omega'/\omega)^2$. (e) Combine: $(dP/d\Omega')/(dP/d\Omega) = (\omega'/\omega)^4$.

(a) The 3–4–5 triangle in S

Length contraction acts only along $\hat{\mathbf{x}}$. The shortest side ($3L$) lies along $\hat{\mathbf{x}}'$:

$$3L \rightarrow 3L/\gamma.$$

The medium side ($4L$) lies perpendicular to the boost (along $\hat{\mathbf{y}}'$):

$$4L \rightarrow 4L.$$

The hypotenuse runs in S from $(3L/\gamma, 0)$ to $(0, 4L)$. Apply Pythagoras:

$$\text{hypotenuse}^2 = (3L/\gamma)^2 + (4L)^2.$$

Take the square root:

$$\text{hypotenuse} = L\sqrt{9/\gamma^2 + 16}.$$

Area in S' (proper frame):

$$A' = \frac{1}{2} \cdot 3L \cdot 4L = 6L^2.$$

Angle of hypotenuse with x -axis in S (acute):

$$\tan \alpha_S = \frac{4L}{3L/\gamma} = \frac{4\gamma}{3}.$$

$$\boxed{3L \rightarrow \frac{3L}{\gamma}, 4L \rightarrow 4L, 5L \rightarrow L\sqrt{16 + 9/\gamma^2}, A' = 6L^2, \tan \alpha_S = \frac{4\gamma}{3}.} \quad (18)$$

Limit $v \rightarrow c$: $\gamma \rightarrow \infty$, triangle collapses to a segment of length $4L$ along $\hat{\mathbf{y}}$.

(b) Wavefront distance invariance

Phase $\Phi = \omega t - \vec{k} \cdot \vec{x}$. Take two points A, B on the same wavefront in S at $t = 0$. Place A at origin:

$$A : (0, \vec{0}), \quad B : (0, \vec{r}).$$

Wavefront geometry requires $\vec{r} \perp \hat{\mathbf{k}}$ (B on same constant-phase surface as A):

$$L^2 = |\vec{r}|^2 = r_x^2 + r_y^2 + r_z^2.$$

Boost along $+\hat{\mathbf{x}}$ to S' and apply the Lorentz transformation to event B :

$$c\Delta t' = \gamma(c \cdot 0 - \beta r_x) = -\gamma\beta r_x.$$

Spatial coordinates of B in S' :

$$\Delta x' = \gamma(r_x - \beta \cdot 0) = \gamma r_x, \quad \Delta y' = r_y, \quad \Delta z' = r_z.$$

In S' , events A and B are not simultaneous, so direct subtraction does not give the wavefront distance. Use the invariant interval $\Delta s^2 = c^2\Delta t'^2 - |\Delta \vec{r}'|^2$. In S :

$$\Delta s^2 = 0 - L^2 = -L^2.$$

Compute Δs^2 in S' from the boosted coordinates:

$$\Delta s^2 = c^2\Delta t'^2 - (\Delta x'^2 + \Delta y'^2 + \Delta z'^2).$$

Substitute the boost results:

$$\Delta s^2 = \gamma^2\beta^2 r_x^2 - \gamma^2 r_x^2 - r_y^2 - r_z^2.$$

Factor the r_x^2 pieces:

$$\Delta s^2 = \gamma^2(\beta^2 - 1)r_x^2 - r_y^2 - r_z^2.$$

Use $\gamma^2(\beta^2 - 1) = -1$:

$$\Delta s^2 = -r_x^2 - r_y^2 - r_z^2 = -L^2.$$

Slide B along the null wavefront in S' to common $t' = 0$; this does not change Δs^2 . At common $t' = 0$ the interval becomes $\Delta s^2 = -L'^2$:

$$\boxed{L'^2 = r_x^2 + r_y^2 + r_z^2 = L^2.} \tag{19}$$

The wavefront-intrinsic distance is the projection onto a null hypersurface and is preserved by Lorentz transformations.

(c) Intensity ratio

Define a flux box of wavefront area A and length λ , containing photon number density n . Intensity (energy per area per time):

$$I = n\hbar\omega c.$$

Total photons in the box, $nV = nA\lambda$, is Lorentz invariant:

$$nA\lambda = n'A'\lambda'.$$

Wavefront area is invariant by (b): $A = A'$:

$$n\lambda = n'\lambda'.$$

Solve for density ratio:

$$\frac{n'}{n} = \frac{\lambda}{\lambda'} = \frac{\omega'}{\omega}.$$

Form the intensity ratio:

$$\frac{I'}{I} = \frac{n'\hbar\omega'c}{n\hbar\omega c} = \frac{n'}{n} \cdot \frac{\omega'}{\omega}.$$

Substitute:

$$\frac{I'}{I} = \frac{\omega'}{\omega} \cdot \frac{\omega'}{\omega}.$$

$$\boxed{\frac{I'}{I} = \left(\frac{\omega'}{\omega}\right)^2.} \quad (20)$$

(d) Aberration and solid-angle ratio

Doppler (given):

$$\omega = \gamma\omega'(1 - \beta \cos \theta'). \quad (21)$$

Inverse Doppler (source frame to observer, observer moves at $-\beta$ in source frame):

$$\omega' = \gamma\omega(1 + \beta \cos \theta).$$

Multiply the two relations:

$$\omega\omega' = \gamma^2\omega\omega'(1 - \beta \cos \theta')(1 + \beta \cos \theta).$$

Divide by $\omega\omega'$:

$$1 = \gamma^2(1 - \beta \cos \theta')(1 + \beta \cos \theta).$$

Solve for $(1 + \beta \cos \theta)$:

$$1 + \beta \cos \theta = \frac{1}{\gamma^2(1 - \beta \cos \theta')}.$$

Subtract 1:

$$\beta \cos \theta = \frac{1}{\gamma^2(1 - \beta \cos \theta')} - 1.$$

Common denominator:

$$\beta \cos \theta = \frac{1 - \gamma^2(1 - \beta \cos \theta')}{\gamma^2(1 - \beta \cos \theta')}.$$

Expand the numerator:

$$\text{numerator} = 1 - \gamma^2 + \gamma^2\beta \cos \theta'.$$

Use $1 - \gamma^2 = -\gamma^2\beta^2$:

$$\text{numerator} = -\gamma^2\beta^2 + \gamma^2\beta \cos \theta' = \gamma^2\beta(\cos \theta' - \beta).$$

Substitute and cancel γ^2 :

$$\beta \cos \theta = \frac{\beta(\cos \theta' - \beta)}{1 - \beta \cos \theta'}.$$

Cancel β :

$$\boxed{\cos \theta = \frac{\cos \theta' - \beta}{1 - \beta \cos \theta'}}. \quad (22)$$

Differentiate with respect to $\cos \theta'$:

$$\frac{d(\cos \theta)}{d(\cos \theta')} = \frac{(1 - \beta \cos \theta') - (\cos \theta' - \beta)(-\beta)}{(1 - \beta \cos \theta')^2}.$$

Expand numerator:

$$\text{numerator} = 1 - \beta \cos \theta' + \beta \cos \theta' - \beta^2 = 1 - \beta^2.$$

Substitute:

$$\frac{d(\cos \theta)}{d(\cos \theta')} = \frac{1 - \beta^2}{(1 - \beta \cos \theta')^2} = \frac{1}{\gamma^2(1 - \beta \cos \theta')^2}.$$

Use $d\Omega = -d(\cos \theta) d\varphi$, $d\varphi = d\varphi'$:

$$\frac{d\Omega}{d\Omega'} = \frac{1}{\gamma^2(1 - \beta \cos \theta')^2}.$$

Identify $\gamma(1 - \beta \cos \theta') = \omega/\omega'$ from (21):

$$\boxed{\frac{d\Omega}{d\Omega'} = \left(\frac{\omega'}{\omega}\right)^2}. \quad (23)$$

(e) Fourth-power beaming

At the source, a wavefront patch of (invariant) area dA carries power $dP = I dA$ into solid angle $d\Omega$. The same patch is seen at $d\Omega'$ in the observer frame:

$$\frac{dP}{d\Omega} = \frac{I dA}{d\Omega}, \quad \frac{dP}{d\Omega'} = \frac{I' dA}{d\Omega'}.$$

Form the ratio:

$$\frac{dP/d\Omega'}{dP/d\Omega} = \frac{I'}{I} \cdot \frac{d\Omega}{d\Omega'}.$$

Substitute (c) $I'/I = (\omega'/\omega)^2$ and (d) $d\Omega/d\Omega' = (\omega'/\omega)^2$:

$$\frac{dP/d\Omega'}{dP/d\Omega} = \left(\frac{\omega'}{\omega}\right)^2 \cdot \left(\frac{\omega'}{\omega}\right)^2.$$

Combine:

$$\boxed{\frac{dP/d\Omega'}{dP/d\Omega} = \left(\frac{\omega'}{\omega}\right)^4}. \quad (24)$$

Question 4 — Constant proper acceleration and the Rindler horizon

Problem. (a) Constant 3-force F_0 from rest: find $v(t), x(t)$ and Newtonian limit. (b) Worldline $x^2 - c^2 t^2 = c^4/a_0^2$: show a_0 is the proper acceleration. (c) Rewrite via $\sinh \omega = a_0 t/c$ in terms of proper time. (d) Photon from $(t, x) = (-D/c, 0)$: find τ_{meet} . (e) Show photons emitted at $t > 0$ never catch up; sketch.

(a) Velocity and displacement

Integrate $dp/dt = F_0$ from rest:

$$p(t) = F_0 t.$$

Use $p = \gamma mv = mv/\sqrt{1 - v^2/c^2}$ and square:

$$p^2(1 - v^2/c^2) = m^2 v^2.$$

Expand:

$$p^2 = m^2 v^2 + p^2 v^2/c^2.$$

Factor v^2 on the right:

$$p^2 = v^2(m^2 + p^2/c^2).$$

Solve for v :

$$v = \frac{pc}{\sqrt{p^2 + m^2 c^2}}.$$

Substitute $p = F_0 t$ and rearrange to the form requested:

$$\boxed{v(t) = \frac{F_0 t/m}{\sqrt{1 + F_0^2 t^2/(m^2 c^2)}}.} \quad (25)$$

Integrate $x(t) = \int_0^t v \, dt'$ with substitution $u = 1 + F_0^2 t'^2/(m^2 c^2)$, $du = 2F_0^2 t'/(m^2 c^2) \, dt'$:

$$\boxed{x(t) = \frac{mc^2}{F_0} \left(\sqrt{1 + \frac{F_0^2 t^2}{m^2 c^2}} - 1 \right).} \quad (26)$$

Non-relativistic limit $F_0 t \ll mc$, expand $\sqrt{1 + \epsilon} \approx 1 + \epsilon/2$:

$$x(t) \approx \frac{mc^2}{F_0} \left(1 + \frac{F_0^2 t^2}{2m^2 c^2} - 1 \right).$$

Cancel and simplify:

$$x(t) \approx \frac{F_0 t^2}{2m}.$$

(b) Identifying a_0 as proper acceleration

Parametrise the hyperbola:

$$x = \frac{c^2}{a_0} \cosh \eta, \quad ct = \frac{c^2}{a_0} \sinh \eta.$$

Differentiate:

$$\frac{dx}{d\eta} = \frac{c^2}{a_0} \sinh \eta, \quad \frac{d(ct)}{d\eta} = \frac{c^2}{a_0} \cosh \eta.$$

Compute the proper-time line element from $ds^2 = c^2 dt^2 - dx^2$:

$$\left(c \frac{dt}{d\eta} \right)^2 - \left(\frac{dx}{d\eta} \right)^2 = \frac{c^4}{a_0^2} (\cosh^2 \eta - \sinh^2 \eta) = \frac{c^4}{a_0^2}.$$

Equate with $c^2(d\tau/d\eta)^2$:

$$c^2 \left(\frac{d\tau}{d\eta} \right)^2 = \frac{c^4}{a_0^2}.$$

Solve and integrate:

$$\frac{d\tau}{d\eta} = \frac{c}{a_0} \Rightarrow \eta = \frac{a_0\tau}{c}.$$

4-velocity $U^\mu = dx^\mu/d\tau$, with $d/d\tau = (a_0/c) d/d\eta$:

$$U^\mu = (c \cosh \eta, c \sinh \eta, 0, 0).$$

4-acceleration $A^\mu = dU^\mu/d\tau$:

$$A^\mu = a_0(\sinh \eta, \cosh \eta, 0, 0).$$

Invariant magnitude:

$$A_\mu A^\mu = a_0^2(\sinh^2 \eta - \cosh^2 \eta) = -a_0^2.$$

Take the square root with the appropriate sign:

$$\sqrt{-A_\mu A^\mu} = a_0.$$

$$\boxed{|\vec{a}|_{\text{rest}} = a_0.} \quad (27)$$

(c) Worldline in proper time

Set $\sinh \omega = a_0 t/c$. Use $\cosh^2 - \sinh^2 = 1$:

$$\cosh \omega = \sqrt{1 + a_0^2 t^2/c^2}.$$

From (a) with $F_0/m \rightarrow a_0$:

$$v(t) = \frac{a_0 t}{\sqrt{1 + a_0^2 t^2/c^2}}.$$

Substitute the hyperbolic parametrisation:

$$v(t) = \frac{c \sinh \omega}{\cosh \omega} = c \tanh \omega.$$

Compute γ :

$$\gamma = \frac{1}{\sqrt{1 - \tanh^2 \omega}} = \cosh \omega.$$

Hence $dt/d\tau = \gamma$, so:

$$d\tau = \frac{dt}{\cosh \omega}.$$

Differentiate $\sinh \omega = a_0 t/c$ to get $dt = (c/a_0) \cosh \omega d\omega$:

$$d\tau = \frac{1}{\cosh \omega} \cdot \frac{c}{a_0} \cosh \omega d\omega = \frac{c}{a_0} d\omega.$$

Integrate from 0 to ω :

$$\tau = \frac{c}{a_0} \omega \Rightarrow \omega = \frac{a_0 \tau}{c}.$$

Substitute back into $ct = (c^2/a_0) \sinh \omega$ and the worldline:

$$\boxed{x(\tau) = \frac{c^2}{a_0} \cosh \frac{a_0 \tau}{c}, \quad t(\tau) = \frac{c}{a_0} \sinh \frac{a_0 \tau}{c}.} \quad (28)$$

Verify the hyperbola identity:

$$x^2 - c^2 t^2 = \frac{c^4}{a_0^2} [\cosh^2(a_0 \tau/c) - \sinh^2(a_0 \tau/c)] = \frac{c^4}{a_0^2}.$$

(d) Photon from $(-D/c, 0)$ catches up

Photon worldline:

$$x_\gamma(t) = ct + D.$$

Ship worldline (future branch):

$$x_{\text{ship}}(t) = \sqrt{c^4/a_0^2 + c^2 t^2}.$$

Equate:

$$ct + D = \sqrt{c^4/a_0^2 + c^2 t^2}.$$

Square both sides:

$$(ct + D)^2 = c^4/a_0^2 + c^2 t^2.$$

Cancel $c^2 t^2$:

$$2ctD + D^2 = c^4/a_0^2.$$

Solve for t :

$$t_{\text{meet}} = \frac{c^4/a_0^2 - D^2}{2cD}.$$

Convert to proper time via $ct = (c^2/a_0) \sinh(a_0\tau/c)$:

$$\sinh \frac{a_0\tau_{\text{meet}}}{c} = \frac{a_0 t_{\text{meet}}}{c} = \frac{1}{2} \left(\frac{c^2}{a_0 D} - \frac{a_0 D}{c^2} \right).$$

Match $\sinh \xi = \frac{1}{2}(e^\xi - e^{-\xi})$ with $e^\xi = c^2/(a_0 D)$:

$$\frac{a_0\tau_{\text{meet}}}{c} = \xi = \ln \frac{c^2}{a_0 D}.$$

Solve for τ_{meet} :

$$\boxed{\tau_{\text{meet}} = \frac{c}{a_0} \ln \frac{c^2}{a_0 D}}. \quad (29)$$

Diverges logarithmically as $D \rightarrow 0^+$: photons fired from the spatial origin at $t \rightarrow 0$ asymptote the ship's late-time light-cone $x = ct$, the Rindler horizon.

(e) Rindler horizon

Photon emitted at $t = t_0 > 0$ from $x = 0$, valid for $t > t_0$:

$$x_\gamma(t) = c(t - t_0).$$

Equate to ship:

$$c(t - t_0) = \sqrt{c^4/a_0^2 + c^2 t^2}.$$

Square both sides (introducing potential spurious solutions):

$$c^2(t - t_0)^2 = c^4/a_0^2 + c^2 t^2.$$

Expand the left side:

$$c^2 t^2 - 2c^2 t t_0 + c^2 t_0^2 = c^4/a_0^2 + c^2 t^2.$$

Cancel $c^2 t^2$:

$$-2c^2 t t_0 + c^2 t_0^2 = c^4/a_0^2.$$

Solve for t :

$$t = \frac{t_0}{2} - \frac{c^2}{2a_0^2 t_0}.$$

Subtract t_0 :

$$t - t_0 = -\frac{t_0}{2} - \frac{c^2}{2a_0^2 t_0}.$$

Both terms are negative for $t_0 > 0$:

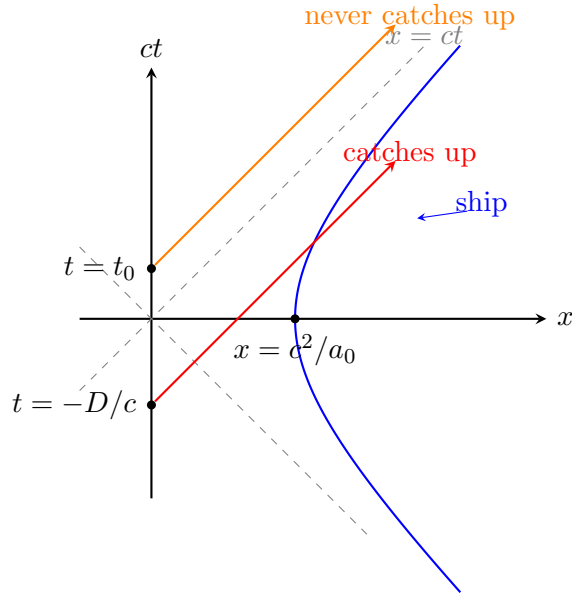
$$t - t_0 < 0 \quad (t_0 > 0).$$

The photon's worldline only exists for $t > t_0$, so the squared equation has no physical solution: no photon emitted at $t_0 > 0$ from $x = 0$ ever reaches the ship.

For $t_0 > 0$, no photon emitted at $(t_0, 0)$ reaches the ship.

 (30)

Spacetime diagram (ship, two photons, horizon):



The orange ray runs forever parallel to the ship's late-time asymptote $x = ct$; the locus $x = ct$ is the Rindler horizon.